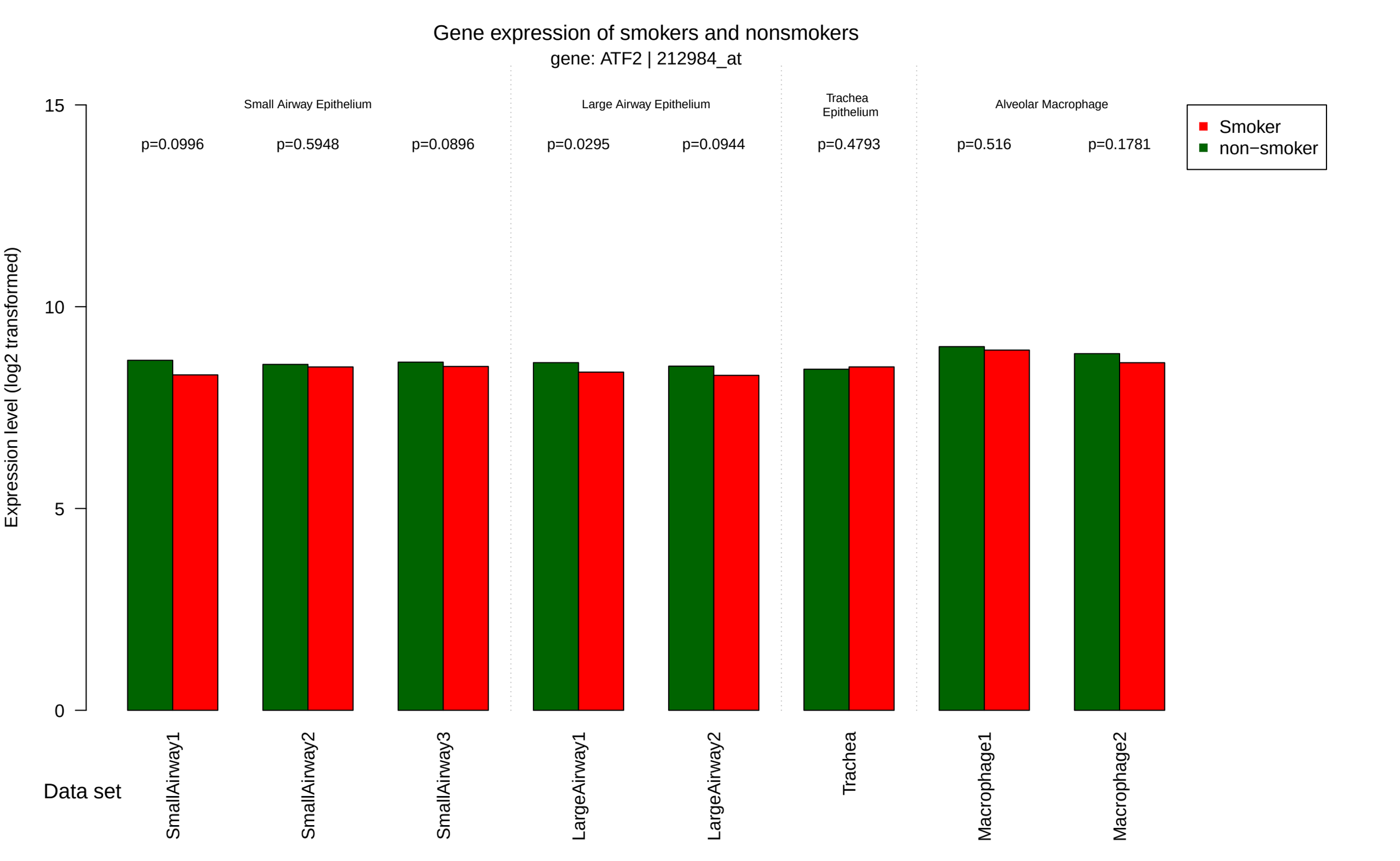
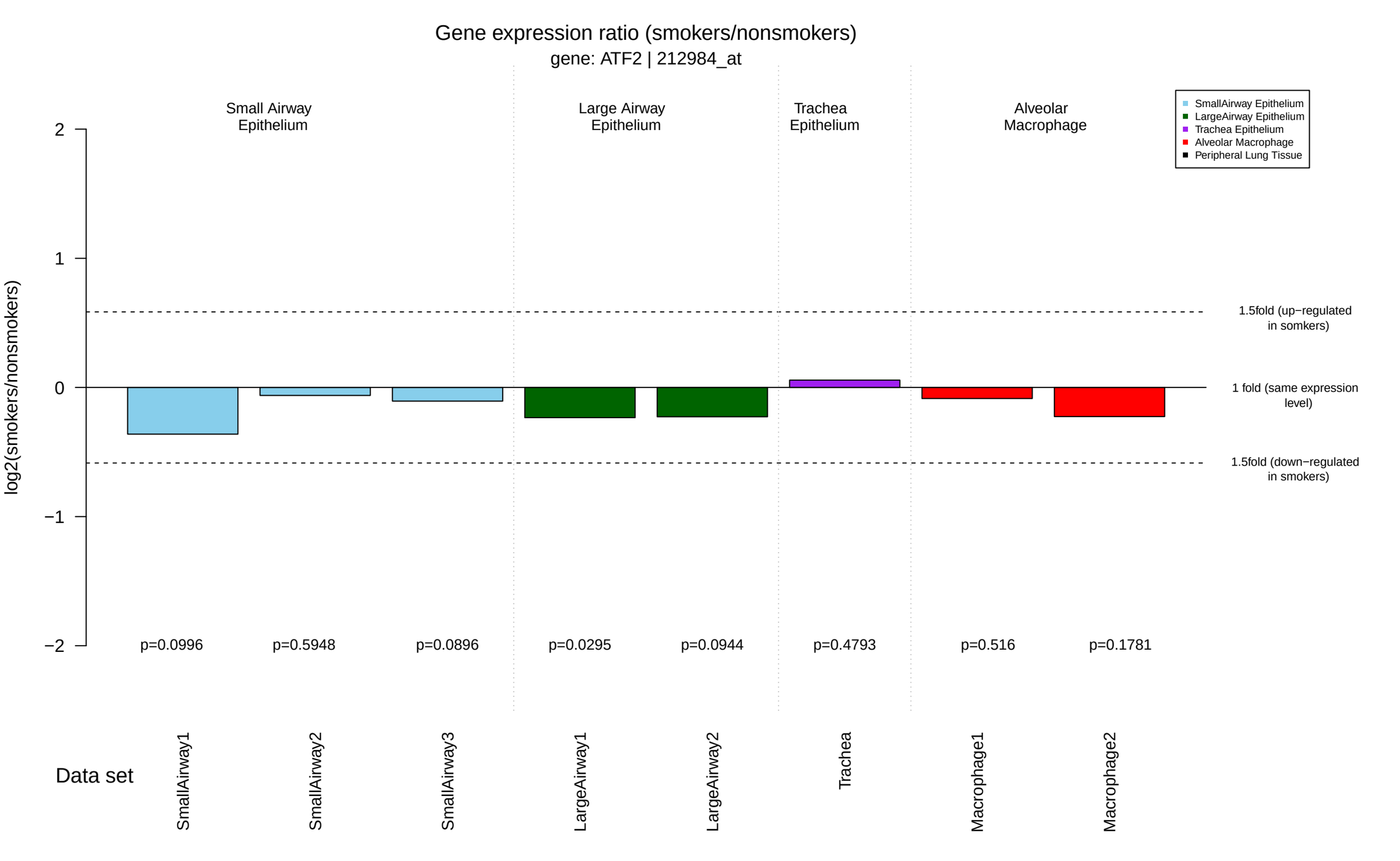
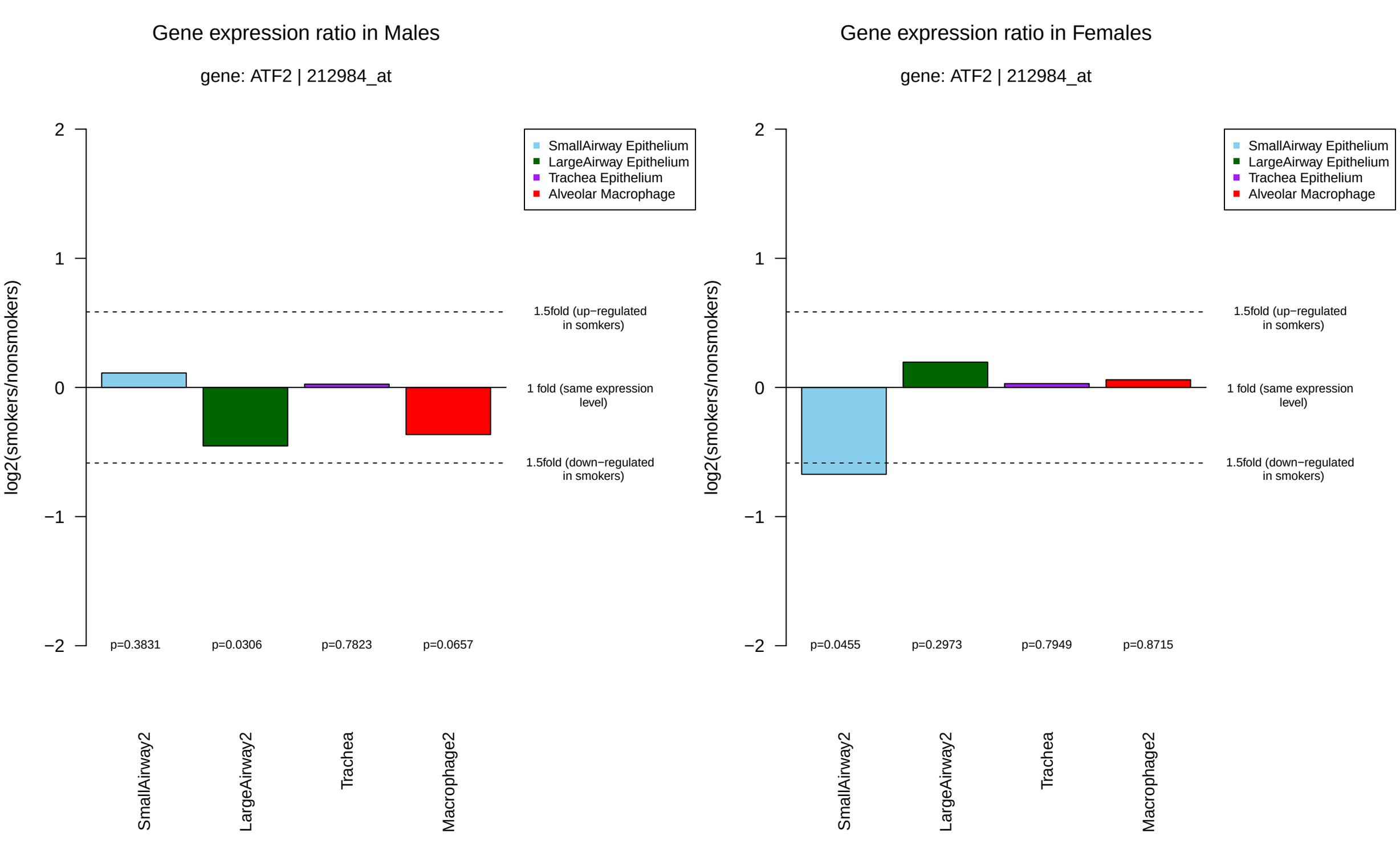
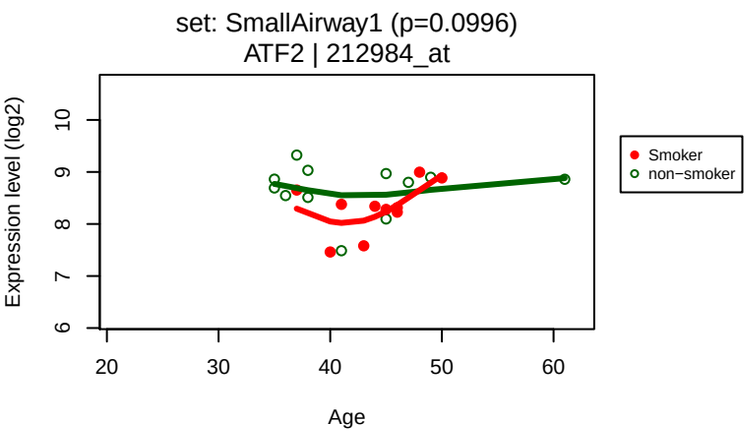
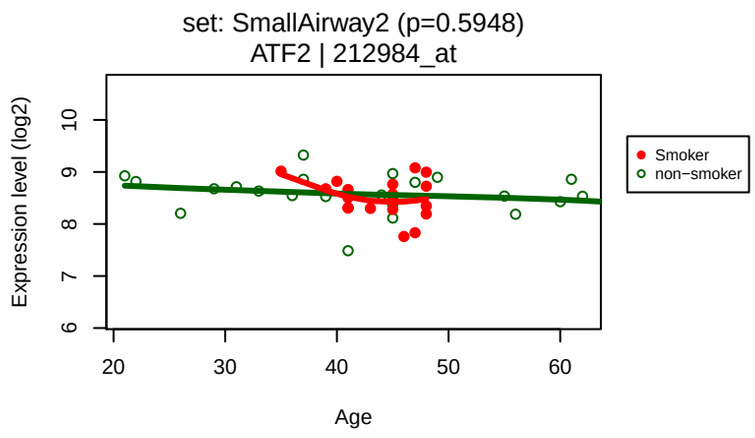
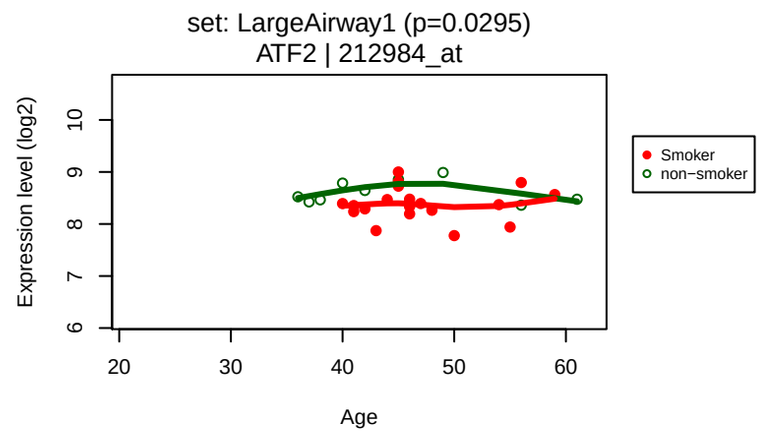

Gene expression of smokers and nonsmokers
gene: ATF2 | 212984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0996
p=0.5948
p=0.0896
p=0.0295
p=0.0944
p=0.4793
p=0.516
p=0.1781
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

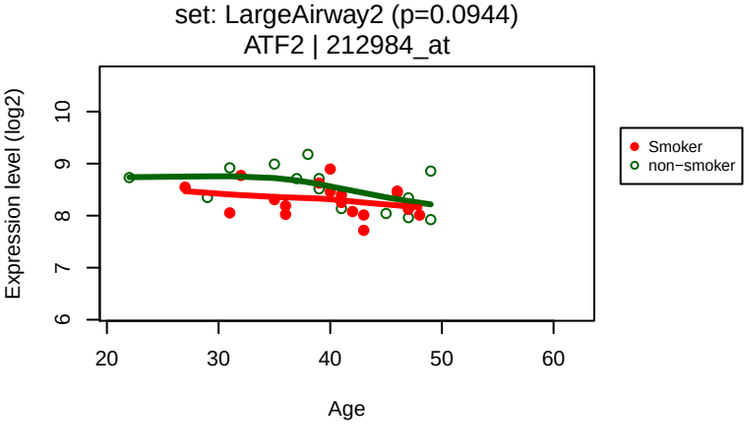

Gene expression ratio (smokers/nonsmokers)
gene: ATF2 | 212984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0996
p=0.5948
p=0.0896
p=0.0295
p=0.0944
p=0.4793
p=0.516
p=0.1781
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

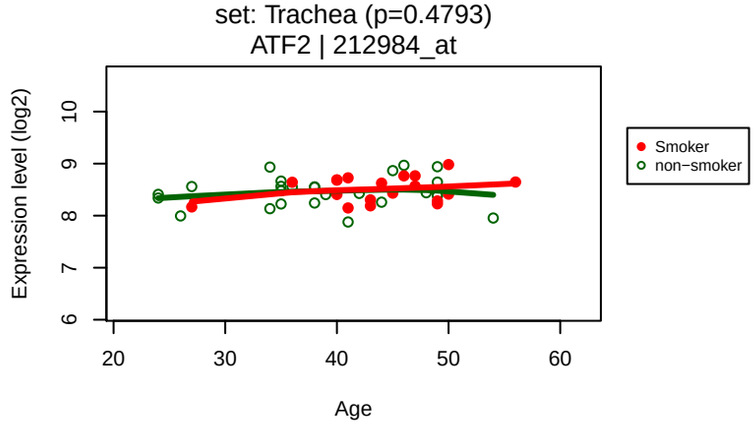

Gene expression ratio in Males
gene: ATF2 | 212984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3831
p=0.0306
p=0.7823
p=0.0657
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF2 | 212984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0455
p=0.2973
p=0.7949
p=0.8715
SmallAirway2
LargeAirway2
Trachea
Macrophage2

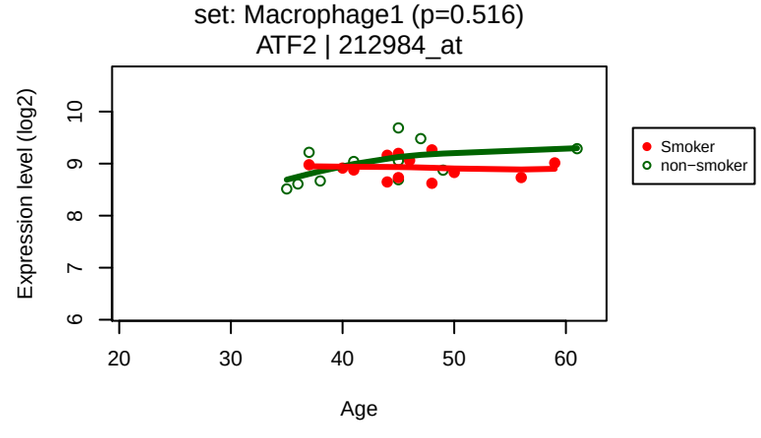

set: SmallAirway1 (p=0.0996)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker

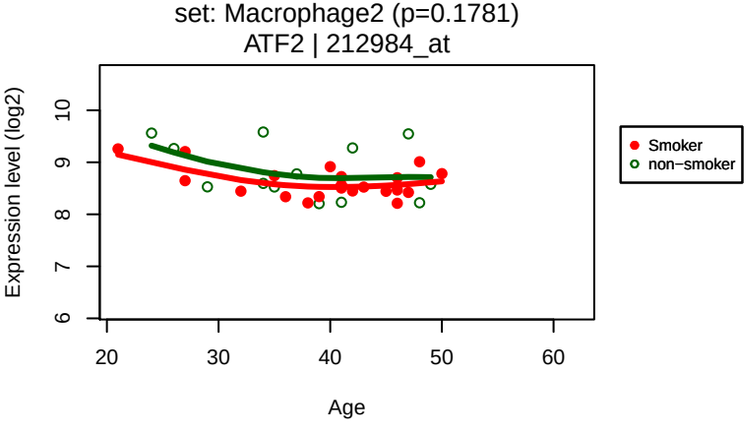

set: SmallAirway2 (p=0.5948)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker

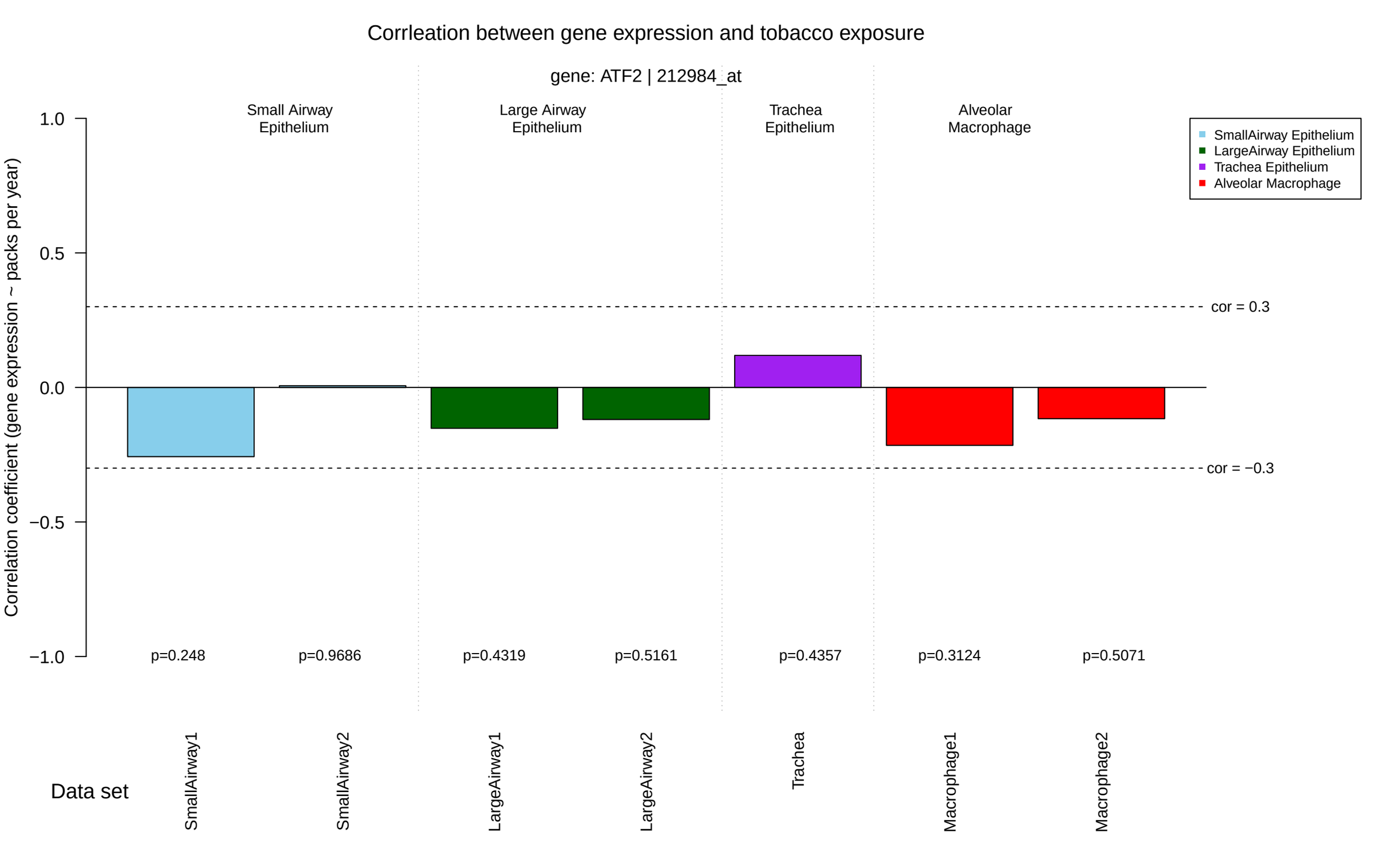

set: LargeAirway1 (p=0.0295)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0944)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4793)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.516)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1781)
ATF2 | 212984_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF2 | 212984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.248
p=0.9686
p=0.4319
p=0.5161
p=0.4357
p=0.3124
p=0.5071
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2